



AI-Driven Solutions Impact On Technical Support Engineers

Xend Finance - Case Study

Joyce Ejezie

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ABSTRACT

Tampereen ammattikorkeakoulu

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JOYCE EJEZIE:

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My thesis looks at how artificial intelligence (AI) is being used in technical support, with Xend Finance as the case study. The research focuses on how AI tools are changing daily workflows, employee responsibilities, and overall efficiency in support operations. Theories that highlight both the technological and organisational factors needed for successful adoption of AI tools in technical support were discussed. I collected data through a questionnaire distributed to 17 employees from different roles within the organisation, including technical support engineers, team leaders, and cross functional staff. The findings indicate that AI is already well embedded in daily operations, particularly in automating repetitive tasks, improving ticket handling, and identifying recurring technical issues. This has allowed engineers to dedicate more time to complex and high value problem solving. However, My study also highlights challenges such as limited trust in AI outputs, insufficient training, and the need for employees to adjust to changing job responsibilities.

Based on these insights, my thesis recommends a structured approach to AI adoption which includes role-specific and practical training, encouraging staff to critically evaluate AI outputs, positioning AI as a supportive partner rather than a replacement, and ensuring continuous leadership support. This study concludes that the most effective adoption takes place when AI and human expertise are balanced, enabling operational efficiency while preserving the central role of human judgment in technical support.

Keywords: Artificial Intelligence, Technical Support, Human and Machine Collaboration.

The Use of AI Tools in This Thesis

- ChatGPT 3.5

During the preparation of this thesis, ChatGPT 3.5 was used as a language support tool to assist with grammar correction, sentence refinement, and improving overall readability. All input provided to the tool originated from my own writing.

The feedback and suggestions generated by the AI were carefully reviewed and edited by me to ensure that the final text accurately represents my original thoughts and academic intentions. No part of the final version contains sentences produced entirely by AI.

I acknowledge that I am fully responsible for all content presented in this thesis, including any sections that benefited from AI assisted language review. I also confirm my commitment to adhere to the ethical principles and academic integrity standards required by TAMK.

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1 INTRODUCTION

Businesses everywhere are working to satisfy and retain customers, build strong brands, and foster client loyalty. To meet these needs in the new era, Technical Support plays a crucial role. The rapid progress of artificial intelligence (AI) has led many scholars to investigate its impact on Technical Support. Du Toit et al. (2020) pointed out: "The impact of the pandemic fast-forwarded this evolution." COVID-19 has reformulated consumer habits and labour practices, driving increasing dependence on multiple types of digital technologies. This change led to an increase in remote support offerings and self-service for customers.

Every day we are creating enormous swathes of data, which can be very difficult to manage efficiently; as Panetta (2016) states, Most of the data is unstructured, and the existing data cannot be effectively used and implemented. In addition, as HBR (2025) states, organisations are accumulating large amounts of data about consumer behaviour across online and offline touchpoints due to the proliferation of digital technologies. Some creative methods have been proposed to improve product and productivity in TS (technical support) environments.

Traditional tech support is plagued by long response times, Ineffective Solutions, and disgruntled customers' problems. AI-driven solutions seek to provide an efficient solution, as we will discover from our case study, Xend Finance. The urgency of change for IT leaders who want to explore new avenues of revenue generation is stressed in ServiceNow (2025), with their view that " the race is on for IT leaders to use emerging technologies like Generative AI (GenAI)". Tech support firms can use these advancements as an aid to improve their operations. Although dialogue about the adoption of AI in Service Management has been ongoing for many years, the recent evolution of Generative AI has created new opportunities to change the way that Support Engineers work in a variety of industries and sectors in which they need to interact well with computer systems. It has contributed to user support and efficient support of technicians for service management.

1.1 Definition Of Terms

1.1.1 AI-driven solutions

Artificial intelligence (AI) driven solutions are advanced technologies that employ algorithms, data analysis, and computing power to automate tasks, predict, and learn from data without the need for explicit programming²⁵⁵.

Jamil Valliani (2024). These are all solutions that are simply more of the same speed, precision, and creativity! AI has transformed how a number of industries do business. AI replicates and incorporates into itself human intelligence through the employment of algorithms and data. These processes are based on machine learning, a process in which computers learn patterns and make decisions without necessarily being programmed to do so in each case.

1.1.2 Tech Support Engineers

These are experts dedicated to handling Information Technology (IT) issues for an organisation's clients and staff. Some work within an organisation, helping employees with system-related problems, while others are more outward-facing, dealing only with clients.

A technical support engineer is responsible for helping users overcome technical problems related to computer networks or systems. Their duties often include troubleshooting issues, guiding users step-by-step through possible solutions via phone, chat, or face-to-face interactions, and keeping track of these cases through support ticket systems. In some companies, the support is focused on internal staff, while in others, it's customer-facing. With the rise of AI, tools such as natural language processing and machine learning are increasingly being used to enhance this support process (Coursera, 2025).

1.1.3 Impact

is seen as "having an influence on something" (Cambridge University Press, 2025). Impact is a complex concept. In this study, the impact is referred to as how the application of these smart tools influences the work of technical support engineers. That includes modifications to what they do, how they do it, how quickly

(or well) they're doing it, how they feel about doing it, what they may need to do it better and whether the jobs they have will remain of value in the future. The effects can be positive (such as easier or faster work) or difficult (such as concerns about losing their job).

1.1.4 Machine Learning (ML)

Machine learning is a technique to instruct a computer to learn from examples. When data is uploaded to, or presented to, a computer, the computer can identify patterns and predict results without being explicitly told how to make those connections. This tool is used for movie recommendations, and social media pages use it to train their algorithm to make predictions based on users' interests. Machine learning is designed to enable computers to act rationally when performing complex tasks. Artificial intelligence systems are adapted to solve problems in a manner similar to how humans would handle the same problems or tasks.

The aim of AI is to develop computer models that can impersonate "intelligent behaviours", as people, explains Boris Katz, a principal research scientist and head of the InfoLab Group at CSAIL. That is, machines that can interpret a visual scene, understand a body of text written in natural language, or take an action in the physical world. Favourite streaming platforms, such as Netflix, YouTube, etc., or to forecast financial data (MIT Technology Review, 2020).

1.1.5 Natural Language Processing (NLP)

NLP enables computers to understand and produce human language. It's what powers virtual assistants like Siri or Alexa to comprehend your commands and answer them in a natural way. NLP also drives language translation tools and sentiment analysis on social media. In reality, "NLP is used in front-end applications for customers or employees that need to be context aware. NLP uses ML to enhance semantic and syntactic recognition. (Lexalytics, 2024).

1.1.6 Xend Finance

This thesis seeks to investigate the influence of AI-based tools on the tasks of technical support engineers at Xend Finance. It will examine the effect of AI on

productivity, workloads, job satisfaction and the potential stumbling blocks for AI adoption. "Xend Finance (RWA), with our global crypto bank offering up to 15% interest, and an open Web3 infrastructure for developers. It makes it easy for its users to obtain fractional ownership of real-world assets, such as real estate, art, and stocks. It makes the world of value creation easy and provides exclusive investing opportunities," (Bigwhale, 2025).

1.2 Research Objectives

The clear and specific statements that define this study are as follows:

To evaluate the advantages and limitations of AI-enabled tools proposed to technical support representatives in Xend Finance.

How job descriptions, skills, and career paths will change in Xend Finance with AI integration.

To make suggestions for optimising AI assistance and the use of human experts as part of tech support.

1.3 Research Questions

The main research question will be answered at the end of the study and will follow:

- How do AI solutions affect support staff at Xend Finance?

The sub-research questions for this study are:

- What are the advantages and challenges of AI-based solutions for technical support engineers at Xend Finance?
- How does AI affect job responsibilities, skills required and career opportunities of a technical support engineer at Xend Finance?
- What suggestions do you have to maximise AI integration, without losing the human touch, in tech support?

1.4 Significance of the Study

The results of this thesis will be of importance to multiple parties who are directly and indirectly associated with the tech sectors and supplying technical support to customers. Stakeholders such as: Managers, Human Resources, Tech support

Engineers, Business Leaders, Developers, Government agencies, etc., can benefit from the result and utilise it to:

- Enhance AI adoption approaches and organisational change.
- Find out how Tech support engineers balance automation with human expertise.
- Discover the results of AI collaboration on Job security and workforce evolution.
- Findings can be used to tailor employee training and upskilling initiatives to the effects of AI.
- To understand the challenges and opportunities in AI-human collaboration towards tech support teams.

1.5 Scope and Limitations of the Study

This study will focus on how innovative tools (chatbots, automation, etc.) are transforming the role of the technical support engineers at Xend Finance. Xend provides digital financial services in various countries. The study targets how such tools impact the daily work of support engineers, what kind of new skills are in demand, how the working speed/quality will be changed and the feeling of using these tools.

Here are the limitations of this thesis :

- This thesis is limited to one company, Xend Finance, and other companies may not demonstrate the same results.
- The company may not share important information, which can constrain the research.
- Because there is limited time allocated to this research, its results may not cover long-term effects.
- Interviewees and respondents can hold back vital information, particularly if they fear for their jobs.

1.6 Research Structure

This structure is the summary of the various chapters of this research. The research objectives lay the foundation for the basis of the research and also outline the intention as well as areas that will be covered in this thesis. This is followed

by the Theory chapter. This will examine existing studies' contributions from renowned scholars that are relevant to understanding this research.

Research methodology will explain the manner, tools and methods of data collection employed in compiling this study. This data will be analysed in the Findings section. Lastly, the discussion aspect of this research will summarise the findings, make recommendations and as well as suggestions for further research.



Figure 1: The research structure.

2 THEORETICAL FRAMEWORK

These theories will be reviewed in order to draw on them in the analysis of this study and to inform the inquiry on the effects of AI solutions on technical support engineers.

The discussion will contain technology adoption theories, organisational change Models, and human-technology interactions to assess how AI integration influences workforce dynamics, operational efficiencies, and overall organisational strategy. We will draw on theories such as the Technology Acceptance Model (TAM), Digital Transformation Theory, Dynamic Capabilities Theory, and Machine Learning Theory to establish a comprehensive structure for understanding AI's role in business operations and work and engagement.

2.1 Technology Acceptance Model (TAM)

The Technology Acceptance Model (TAM) is an information systems theory that describes how users come to accept and use new technologies. Developed in the late 1980s by Fred Davis, predominantly in response to the failure to adopt technology by people or organisations. The outcomes did not align with what was expected, as this model attempts to predict user acceptance by the features of a technological system.

The core concepts of TAM revolve around two key elements: perceived ease of use (the ease of use that potential users consider when using a new technology) and perceived usefulness (the benefit an individual perceives from adopting a technology to perform a task)

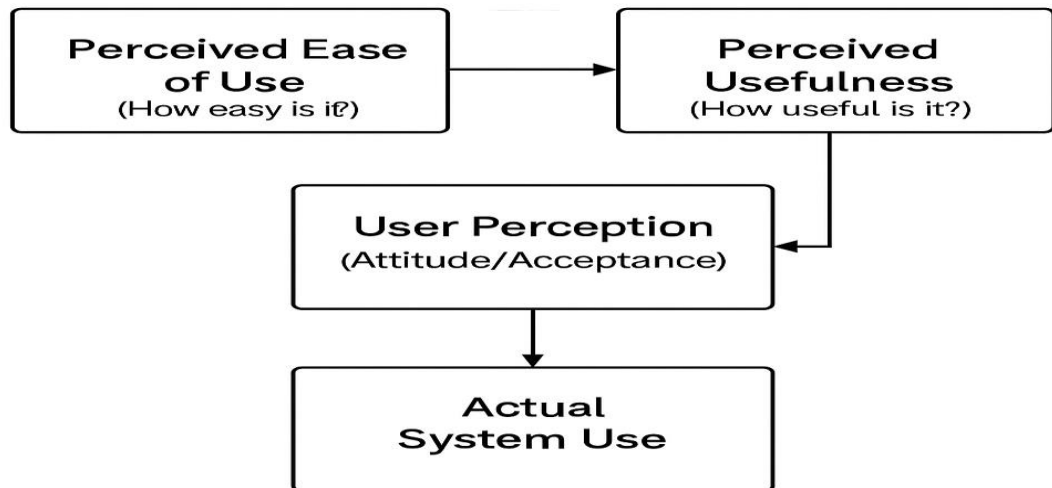


Figure 1. Technology Acceptance Model (TAM). Adapted from ResearchGate (2022).

Together, these components also form user perceptions towards the usage of the system, which significantly impacts their acceptance level. The model has been widely adopted and adapted in several domains by researchers to validate its applicability in terms of explaining the interrelation between technology use and human behaviour (something that is particularly relevant in today's fast-paced technology environment in both home and workplace). (Dziak, 2024).

TAM was originally developed by Fred Davis to explain computer usage and is intended to be field-independent as it is believed this form of analysis works for any usage context, and that predictive value is obtainable by looking at several vital determinants: That is,

- The nature of the technology,
- its features, and
- Operational design aspects.

This elicits a response from the user applying it. Based on these aspects, the user then determines if he likes the system and how much motivated he is to use it. TAM suggests that people will be more likely to accept a new system if they believe it is useful and easy to use. The TAM model captures the design characteristics of a system (functionality, ease of use, benefits). So if, for example, a new accounting software tool can help make computations faster, keep data safe

and aid in analysis, those features are going to affect how users feel about it and whether users even use it at all.

TAM is relevant for this thesis as it helps understand how technical support engineers adopt the AI tools in their day-to-day job. It's worth knowing how they are feeling about these tools. And do they find the tools useful? Are the tools easy to use? Those are the questions that will guide us when it comes to whether the technology will be adopted and used by engineers. For a lot of companies today, AI is helping in things like ticketing triage, intercepting problems before they escalate, and finding the person who can solve a problem the fastest. However, if engineers are not made comfortable with these tools or don't have faith in them, they don't use them very well. "It is not just about the technology when it comes to introducing AI in the workplace, but it's also about how people feel about using it" (Dwivedi et al., 2019).

Usefulness: If the support engineers perceive that an AI tool is making their life easier (maybe helping them to respond to tickets more quickly, or make fewer mistakes), they're more likely to use it. "When people have faith that it benefits them, that's when they're going to use a system" (Legris et al., 2003).

Accessibility: How relevant is it, how convenient is it, or how understandable is it? When it's confusing or too much trouble, engineers may just stay away. "If it's easy to use, people think, 'That's a good idea,' and they are more likely to use it" (Venkatesh & Davis, 2000). "for AI to play a good supporting role, it has to be something people will trust, and they have to know how it's making decisions" (Glikson & Woolley, 2020).

For technical support applications, AI is used to predict how to solve customer issues through referring to logs to notice issues, to send alerts of likely problems or discrepancies, and to prioritise which tickets should be solved first.

However, when and if such tools or AI apps are effective, easy to use, and assist engineers, they will utilise them as part of their job. But if they get too complicated, they might be ignored entirely. It also helps us understand why people won't use AI. Some common reasons include: Not trusting the tool's suggestions, Misconceptions on tool use not getting enough training, poor design and difficult to work with the tools "AI must be both trusted and understood by the people it is designed

to support in order to help them perform their jobs more effectively,” concluded Glikson and Woolley (2020).

TAM has demonstrated significant success in other fields, such as health, and in the context of AI in customer support. For example, Alalwan et al. (2018) investigated AI chatbots and found that trust and perceived ease of use significantly influenced people’s acceptance of them. Liu et al. (2021) studied AI tools used at IT help desks and found that training significantly increased the likelihood of users adopting the tools. These findings, along with similar studies, suggest that when engineers are properly trained and the tools are transparent and trustworthy, the acceptance and use of AI solutions becomes more likely.

2.2 Digital Transformation Theory

Digital Transformation theory was advanced by Westerman, Bonnet & McAfee (2014) and "It means the companies need to become what we call digital masters. They do this by building two types of capability: digital capability (the ability to use new technologies to access and engage different parts of their business) and leadership capability (the ability to envision and lead the kind of organisational changes that are necessary to capture the digital opportunity in thoughtful, and profitable, ways) and together, they allow a company to transform digital technology into business advantage."

That is the notion that companies will integrate digital technologies (AI, the cloud, automation) to make their operations run faster and more efficiently.

Globally, most business organisations have embraced and reimplemented modern technology to rewire the way they run their business and serve their clients. It's this notion of digital transformation, not just adding shiny new tools/technologies. It's one that, ultimately, represents a deeper, more strategic change to business model, organisational culture and operational capabilities.

(Westerman, Bonnet, and McAfee 2014) publicise the need for sophisticated digital tools to be married with strong management to produce significant results. And digital masters are, as a group, more effective with technology than laggards at creating new value for customers and enhancing operational efficiency.

This theory is relevant for the interpretation of the AI-assisted tool's effects on the job of a technical support engineer. Here, Digital Transformation Theory helps to explain why businesses are gradually adopting AI in technical support processes. AI is not a technology for its own sake, and AI is being deployed as part of a wider strategy to increase efficiency, minimise the likelihood of human error, and create a better customer experience.

Here, support engineers, during this shift, are told we need to alter how we do our job, our job responsibilities. Where once humans were directed to diagnose customer issues, AI systems in the future can be focused on analysing system logs, suggesting what likely causes might be, and possibly generating what could be recommended solutions. All these changes can bring efficiency, but there are new requirements put upon the humans, who now need to retrain, relearn, and trust the new AI tools that are provided to them.

As (Bharadwaj et al. 2013) contend, the transition to digital is not just an investment in new technologies, but a move to transform the strategic mindset of the organization. It demands that companies be organised in a manner which enables their people, processes, and technology to embrace continuous innovation. This is particularly the case in technical support teams, as employees are required to be trained to operate not only AI tools, but also how these actually work and their limitations.

The impact that digital transformation has had on support engineers has been both daunting and empowering. On the one hand, mechanical jobs requiring a lot of manpower, in which you had to spend time tracking through tickets, finding issues that have come up time and time again, or prioritising workloads, can now be automated by AI. This allows engineers to use their time for deep, complex, high-level thinking, which might even be more fun and less repetitive.

But it's not without its own set of expectations. Engineers can't just get good at using AI tools; they need to learn to trust AI recommendations (or not), assess the extent to which AI algorithms get things right, challenge them when they're wrong, and not be ashamed to speak up. The role of the engineer shifts from simply using tools to interleaving with the AI-enabled support process.

(Hess et al. 2016) offer an important contribution in this direction. They argue that an organisation's successful passing through digital transformation is primarily determined by the level of support it provides its employees through the process. This means providing the right training, being upfront with the expectations of changes, and providing tools that not only work but are also easy to understand and use. "It is critical for the accomplishment of digitalisation that an organisation is capable of adapting to new digital technologies and that employees use them for work purposes" (Hess et al., 2016).

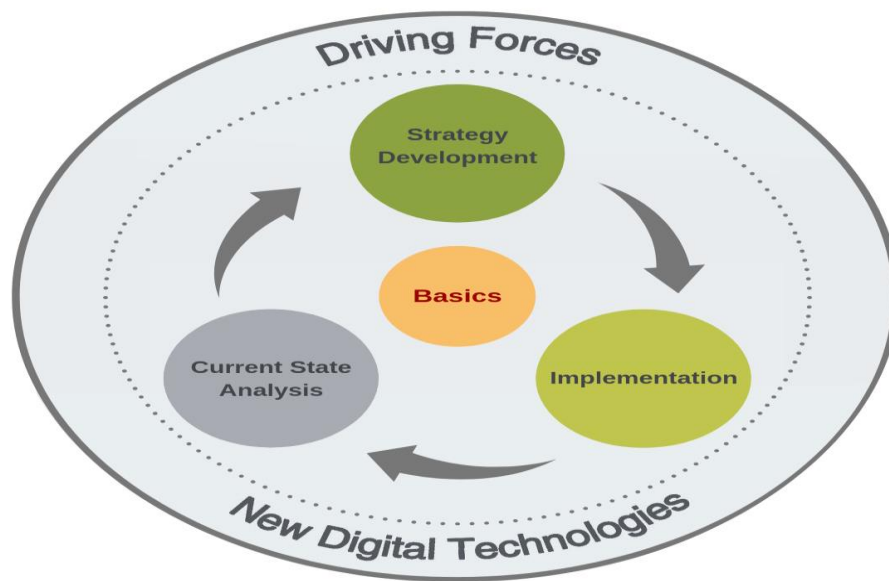


Figure 3: Digital Transformation Learning Adapted from Digitalisation project (2025)

According to the image, digital transformation is a (repeatable) cycle. For that to work in technical support environments, however, organisations need to spend as much effort on people management and change leadership as they do on the tools alone. In the middle is "Basics," readiness for change at the core. This includes the mindset of employees, digital savvy, and a grasp of business.

The first significant move is Current State Analysis, the process in which the company evaluates its current solutions and issues. For technical support engineers, that means investigating their pain points before implementing AI. The next stage, Strategy Development, will focus on planning how digital tools, such as those powered by artificial intelligence, will be employed. The Tactics should be in relation with the objectives of the Business and the capabilities of the employee.

Next comes Implementation, where AI utilities are added to everyday tasks. Support engineers start taking advantage of features such as automated responses or smart routing. Good training and support are key to succeeding at this stage. In the wake of an implementation, the loop returns to a consideration of what is now here. This is a cycle of feedback that permits learning and adaptation. Driving Forces (e.g. Customer Needs, Competitive pressure) surround the cycle. These dynamics force organisations to innovate and reinvent.

On the far outside are New Digital Technologies, such as AI, that permit transformation. The theory focuses on the point that digital transformation cannot be a one-time affair. It's a process of strategy, execution, learning and organisational change. This model shows how AI is rolled out to support engineers through structured change in the organisation.

2.3 Dynamic Capabilities Theory (DCT)

It was developed by Teece, Pisano & Shuen (1997) and elaborated by Teece (2007, 2018). "Dynamic capabilities can be contrasted with operational or 'ordinary' capabilities, which are those that organisations use to compete in the present marketplace. Dynamic capabilities, by contrast, "are the firm's capacity to create, develop or modify its resource base purposefully, " as defined by (Helfat et al. 2007). The underlying premise of the dynamic capabilities framework is that core competencies can be utilised to adjust near-term competitive positions that, in turn, help to underpin long-term competitive advantage. (Shuen, 1997) pioneered this idea by emphasising the role of managerial and organisational processes for achieving competitive advantage.

The premise is that, in order to remain in the market and to remain in the lead position, every company must, on a continual basis, locate its opportunities, capture them, and then allocate and re-allocate resources in such a way that the company can seize those opportunities. Its also revolves around the company's ability to put that change into practice based on the ongoing training they receive to acquire new skills, retrain old processes, and get to work with new technologies, including AI.

The essence of this theory is that organisations cannot afford to be good only in what they are doing. They can no longer ignore, unlearn, or be non-adaptive. These competencies are what academics refer to as "dynamic capabilities." They go well beyond basic day-to-day activities and extend to skills capable of sensing changes in the market, adjusting strategies, and re-shaping internal processes. What this comes down to is reading the environment correctly, deciding what needs to be changed, and then making the necessary adjustments. This may involve learning new things, investing in people, and even rethinking the way the company operates in some cases.

Teece (2007) developed the theory further by defining three fundamental dynamic capabilities as follows:

- Sensing: the capability of recognition of opportunities and threats.
- Seizing: the capacity to mobilise resources to appropriate value from opportunities.
- Reconfiguring: the capacity to change and redevelop their capabilities as necessary.

This theory is crucial in explaining how AI tools will impact the work of technical support engineers and how they integrate AI tools successfully. To attain this, the organisation need to be included in the process and they are responsible for the following:

Anticipating Change: No business wants to be left behind. It's important to stay aligned with what customers expect, like quick answers and the ability to solve problems on their own. Tools like chatbots and predictive diagnostics can help meet those expectations. But before any of that works, the company has to genuinely recognise the value these technologies bring.

Sensing this change requires scanning, searching, and exploring across technologies and markets. That's being proactive about change, not just reacting when things go wrong. (Teece, 2007). Any company that fails to recognise the growing importance of AI in support functions risks falling behind, especially as competitors are already using these tools to their advantage.

Seizing the Opportunity: This comes next. When a company discovers a way that AI can help its support engineers, it must put the proper resources in the form

of tools and training into the project. This step is called "seizing." It's a matter of making abstract opportunities. In practice, this means investing in software that employs machine learning to recognise your customers' most common problems. But more important is the training it provides the engineers to understand the system and its recommendations, so people will actually use it in their daily work." Seizing involves commitments of time, resources, and investment in such activities as the acquisition of new skills or the reshaping of current business models" (Teece, 2007). Dynamic capabilities are as much about people and skills as they are about tools and technologies.

Transforming the Organisation: The third part of the theory has to do with change. The best AI tools aren't going to have much of an impact if the company keeps doing things the old way. That's why forward-thinking companies examine their work processes regularly and make adjustments where necessary. These changes might involve altering the job of a support engineer. Instead of processing every user service request by hand, they may now concentrate on the more complex cases that AI cannot resolve. They might also have more time to refine the AI tool by offering feedback on how well it works. As (Eisenhardt & Martin 2000, 1107) explain, "organisations need to constantly renew and reconfigure their resources to maintain competitiveness".

Change is not an event; it's something that happens all the time. As (Barreto 2010) notes, "developing dynamic capabilities requires deliberate effort, investment in human capital, and a leadership mindset that values agility."

The Dynamic Capabilities Theory explains why some firms achieve success with AI and others don't. And companies with strong dynamic capabilities can identify promising technologies early and are committed to investing in people and systems. They are able to change their organisational design and culture to facilitate new ways of working. These companies don't just purchase AI tools; they put them to work for their businesses. Companies with poor dynamic capabilities, on the other hand, will purchase the same tools, yet go nowhere, since they are unfit for change.

2.4 Machine Learning Theory

Artificial Intelligence Systems are used to solve complex tasks to mimic how a human thinks and approaches a problem. Machine Learning (ML) is an emerging science that enables the use of computers to learn from data and make decisions without being explicitly told what to do at all times. This premise of learning from experience is nothing new to humans, but when applied to machines, it unlocks a world of possibilities, particularly in activities such as technical support. For somebody like me who is working as a support engineer, knowledge of how ML operates and how it comes into play in my routine is crucial. ML is what underlies much of the AI that is applied today in customer service platforms, helpdesk tools and automation systems.

(Blum, 2018) clarified that “A main goal of Machine Learning Theory (also known as Computational Learning Theory) is to understand the basic principles of machine learning by viewing learning as a computation process. This field aims to study at a mathematical level what specifically is needed in terms of capabilities and information to learn different kinds of tasks, and what the fundamental principles of algorithms are to make computers learn from data and get better with feedback. The objectives of this theory are in planning for better learning and developing grounded explanations of how learning occurs, both for the automatic learning of learning methods, as well as for the process of learning itself.”

Machine learning theory is a jumble of ideas, Blum clarified, from how machines work and how we understand data using statistics. It's about determining how we can make machines learn better, using logic and numbers. This is important in technical support because AI tools are increasingly being used, and we want to know if they really learn and assist well.

Creating Models of Learning: One aspiration is to devise simple formulas or systems that describe how learning behaves. These models can help us comprehend which tech support issues are simple enough for AI tools to answer and which ones prove more complicated. Example: An AI tool might just as quickly

figure out how to answer "How do I reset my password?" but have a hard time determining unique errors from custom software installs. A model helps us see why.

Knowing What Works (and When): The theory can also help confirm whether the AI is doing a good job. It raises questions, for example, about when is this AI tool is going to give the right answer. How much training (data and time) must it undergo before it is useful? Example: If the AI chatbot is good at answering questions on tickets after reading 5,000 solved cases, theory can help to estimate how long it needs to learn. It will help teams to properly plan.

Understanding Big Picture Questions: Blum also discusses big ideas, like: Why is picking simpler answers (Occam's Razor) better for machine learning? When does a machine seem trustworthy enough to give good answers based on just a few illustrations? To do this, is it better for the AI to engage (ask questions, receive feedback) or to sit quietly and take in the scene? Can it still learn effectively in situations where much of the data is noisy or irrelevant? Example: In a support role, AI could be misled by long email threads with other (unrelated) customer chatter. But if it can figure out how to focus only on the part that describes the problem, that's smart learning. Machine Learning Theory helps you to understand how that occurs.

Also, the Theory of Machine learning provides our means to construct, trials and understand AI systems. It helps teams trying to support the tools determine what AIs to trust, how much data to feed them and what to expect in return. Machine learning, a subset of artificial intelligence (AI), is the process of using algorithms to make decisions and predictions based on data, without being programmed with specific instructions. Rather than programming a computer to follow a set of fixed rules, machine learning is training the computer to produce predictions or decisions by allowing it to analyse large amounts of data and find patterns on its own.

(Samuel, 1959), One of the pioneers of research in this field wrote that machine learning is a field of study that gives computers the ability to learn without being explicitly programmed. That is, over time, the system can learn to do a better job on a task. The Three Types of Machine Learning are: Supervised (where the machine is fed both inputs and the desired outputs), unsupervised (where it looks

for patterns), and reinforcement (the system learns through reward/penalty over time).

(Liu et al. 2021) described how helpdesk operations are enhanced through machine learning's ability to automate routine tasks as well as assist engineers in making difficult decisions. Their study demonstrated that support engineers using AI-based tools can take on more cases, commit fewer errors, and concentrate on addressing harder problems rather than performing manual work. Many current support applications, such as ServiceNow, Zendesk, and Microsoft Dynamics, are machine learning platforms that power their respective data centres. These tools can also predict what type of problem a user is raising, suggest potential solutions and, at their best, direct issues to the right department. Just as it trains algorithms to tell the simple from the difficult, machine learning is commonly used in customer service for interpreting human language (an activity commonly called natural language processing or NLP) and therefore understand emails, chat or support tickets that have been written in various ways (Lexalytics, 2024).

Also, ML-based tools can be trained on historical support logs to find out the trends. For instance, when a new software update is rolled out and starts to cause bugs, the ML system can put a flag on this pattern sooner than a human might observe it and help engineers to respond more quickly. This is relevant to analyse the effect of AI-assisted engineering tools because AI tools help teams work more efficiently by speeding up repetitive tasks and reducing mistakes. They also support better decision-making by offering suggestions based on past experiences and data. This saves time, allowing engineers to concentrate on more complex and strategic problems while the AI handles the routine work. Over time, working with these tools also helps support engineers grow; they're not just fixing issues manually, but learning new skills as they collaborate with intelligent systems.

(Glikson and Woolley 2020) also discovered that well-designed AI tools can lead to increased trust and confidence in the system among engineers. But the study also cautioned that ambiguous or overly complicated systems could be frustrating and undermine confidence. However, even though the uptake of ML for service roles has its strengths, it is not without its challenges. Automation of this type might make some engineers jumpy; they may imagine it's a threat to their job. For

others, new tools can be too much of a learning curve. As a result, organisations need to help engineers adapt to these tools through training and support.

(Barreto 2010) even makes this point, underlining how the technological changes demand a flexible workforce, inclined to learn and to innovate. That is why it is so critical for companies integrating machine learning tools to have engineers working on this and to make sure the AI supports human work rather than automating it away.

The underlying concept is that ML makes support engineers more capable and decreases time spent on mindless tasks, enabling them to take on more interesting and challenging work. Machine learning affects the support team and customer experience; it might redefine the expectations and necessary skills of a support engineer. ML is driven by the principles of lifelong learning, problem-solving, and adaptability, which are the hallmarks of digital transformation at the workplace.

3 RESEARCH METHODOLOGY

3.1 Data Collection Methods

This study uses two main sources of information, data collected directly from the case company and background material from earlier research on how AI is changing technical support. For this research, we'll be using questionnaires only to gather data from participants in the Xend Finance company. The questions will be shared through Google Forms, making it easy for participants to respond. Once we collect the answers, we'll look for common themes and patterns to help us understand the effects of AI in their support processes and work.

3.2 Research Ethics

Research ethics was considered in the study. All participants will be made aware of the objectives of the study, how their information would be used, and their right to withdraw at any time. The wishes of respondents to maintain anonymity were acknowledged while drafting the questionnaires. They responded anonymously and were not identified by name in the final report. Data was stored on a secure server, and only the researcher had access to it. This study adhered to TAMK 2020 ethical guidance.

3.3 Case Study Xend Finance

"Xend Finance is creating a DeFi tool and high-interest savings for cooperatives and credit unions via blockchain" (Xend Finance Whitepaper, 2020). Xend Finance is a fintech firm operating on the blockchain infrastructure and provides services in the less privileged markets. Xend delivers value to its customers by solving the credit union pain points, such as lack of liquidity and low interest rates, but with superior solutions on a global blockchain network.

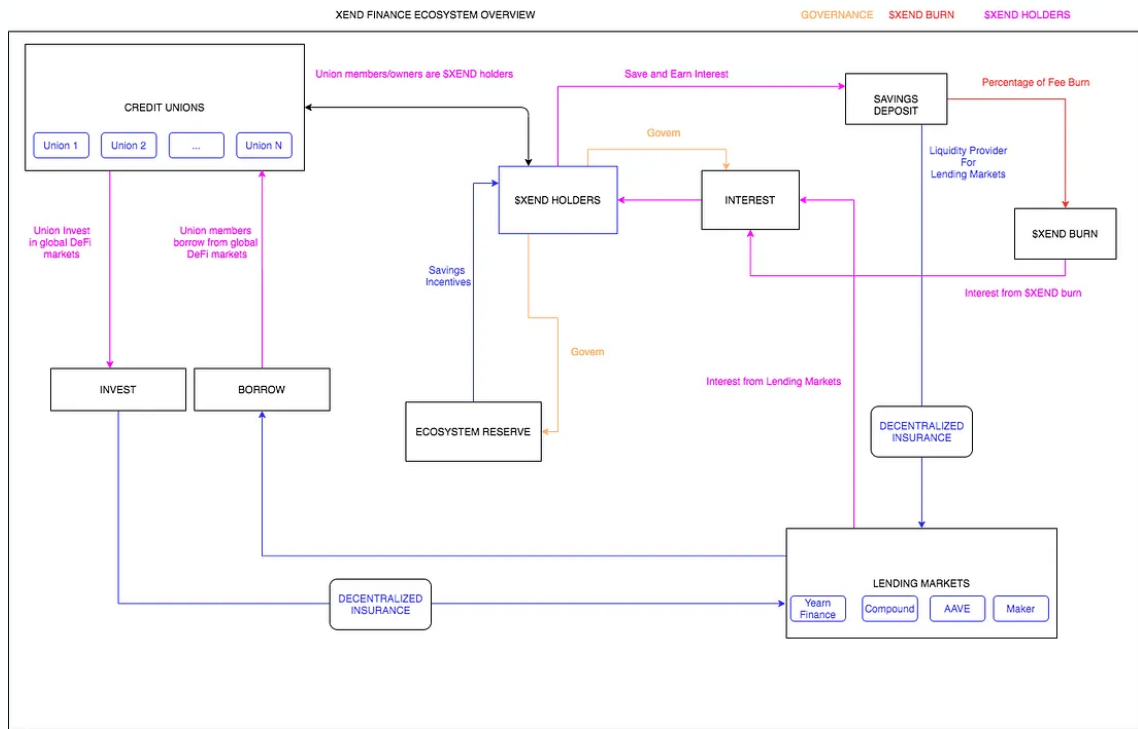


Figure 4: Block Diagram for Xend Finance adapted from XendFinance (2020).

The vision and purpose of Xend is to eliminate geographical barriers by allowing people anywhere in the world to join a cooperative or credit union. The interest rates are also much higher (approximately 15%–20%) compared to the traditional 1%. (Aronu Ugochukwu, 2020). Xend's Finance also secures assets with decentralised insurance and clear smart-contracts.

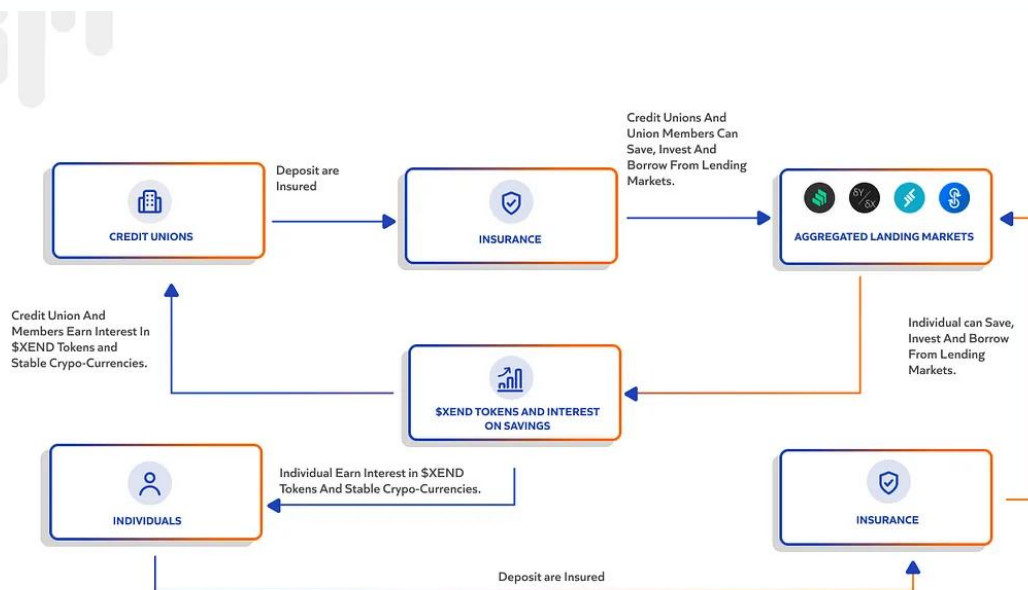


Figure 5: Xend cross-chain credit union. Adapted from: Xend Finance 2025.

Xend leverages two popular savings systems: credit unions and Esusu. These are methods people have used to pool money for many years. Credit unions let people put their money together and take turns to use it or earn interest. Esusu is a rotating savings system where one person gets the money each cycle. What makes Xend different is that it uses smart contracts (automated programs) to manage these group savings, making the process fair and automatic (Xend Finance, 2023).

Earning in Xend has two levels of Interest, and users can earn in two ways:

- Base interest from staking money (usually stablecoins) on DeFi platforms like Aave or Compound.
- Reward tokens, called Xend, are given as bonuses for using the platform.

This way, people get not only routine savings interest but DeFi returns as well as token rewards (AsiaCryptoToday, 2023).

Xend Token is not just a bonus. It is a currency used to pay for saving group services, to vote on platform decisions (with a DAO system) and to get rewards from customers. That is, users have a say in how Xend Finance should evolve or what it should look like in future (CoinGape, 2023).

Xend Technology and Tools were built on Ethereum but shifted to Binance Smart Chain because it's cheaper and faster. They even built a few useful tools like:

- Asset Chain: Their own blockchain for converting physical assets to digital ones.
- SwitchWallet: allows users to interact with various crypto networks such as Bitcoin or Ethereum.
- XendBridge: aids in converting fiat to digital and digital back to fiat.
- MadWallet: Easy and safe to store various tokens in one wallet.

These financial instruments are used to be able to appraise savings and digital assets in one ecosystem (Xend Finance, 2023).

Who Is Using Xend services?

Xend Finance is tailored to Africa, where a significant portion of people lack bank accounts. In 2023, it boasted over 150,000 users. It imitates saving models that the people in countries such as Nigeria, Ghana, and Kenya have believed in so well, and the user feels safe using it (TechInAfrica, 2023). Xend Finance has

raised approximately \$2 million. It received backing from major companies, including Binance and Google Launchpad. This capital helped develop its tools and expand its community (Tracxn, 2023).

Xend Finance's app was released in 2022 and includes features such as:

- Converting local currency into stablecoins (BUSD, for example)
- Flexible or restricted savings plans (returns up to 15%)
- A referral system to earn bonuses by inviting others

It is designed to be user-friendly for everyone to use, even if they are not very familiar with the crypto or web 3 space (TechPoint Africa, 2023).

Xend offers its users:

- Hedge against inflation through the use of stablecoins
- Rising demand from savvy DeFi picks
- Optional insurance on assets
- Financial education and support for low-tech users

These are the things that make Xend a better alternative for people within those regions, where traditional banking doesn't work very efficiently (Medium, 2023).

Inasmuch as Xend has established revolutionary consumer-focused services, there are still some downsides to using Xend:

- The volatility of Crypto markets and blockchain assets.
- Many people are new to blockchain
- There are competing platforms with similar services

For Xend to continue to grow, it must become more trusted, easy to use and secure (Whitepaper, Xend, 2023).

Xend has established a solid community on Telegram, Discord and other channels. It also works with other blockchain networks to bring in further ideas and keep improving. This user-centric methodology fosters trust and encourages innovation (Reddit, 2023). Xend plans to refactor Asset Chain to enable real-world investment (real estate), establish a legal asset tracking system (Global Ownership Register), create a market and social hub for digital markets and provide enabling tools for developers and local agents to go global (Xend Finance, 2023).

Plus, the service has a group of support engineers who assist users with technical issues, bugs and usage questions. Xend Finance has leveraged AI tools to assist its engineers. These tools focus on reducing response time, minimising human error, and increasing satisfaction. Therefore, using Xend Finance as a case study allows us to observe AI implementation in a real-life support environment and understand its direct impact on workers.

3.4 AI Software To Be Used by Engineers In Xend Finance.

Xend Finance Integrates AI Tools To Make Their Technical Support Process More Efficient And User-Friendly. These are tools that assist the support engineers to speed up on generic processes and provide faster resolution to customers. Instead of handling all those sorts of things by hand, engineers now have digital tools that can help with many of the routine requests, allowing them to concentrate on the more challenging questions. (Xend Finance, 2025).

At Xend Finance, one of the important milestones achieved in the area of enhancing customer support has been the development of Xend AI, which is an AI-powered chatbot. This tool plays a crucial role in streamlining communication between the platform and its users. For most people who are looking for assistance or have questions about the service, Xend AI is the first line of contact. It's designed for simple, basic processes, the kind of routine work that doesn't need the input of a live human support agent. For example, the chatbot can instantly report on the status of a transaction or frequently asked questions, as well as assist users in a clear path through the platform.

By dealing with these routine questions, Xend AI alleviates some of the support burden on engineers and customer support agents. This can be especially useful in peak user times, as traffic can often become too much for a database to handle. But the chatbot is not meant to replace human aid completely. For more challenging queries or when a customer needs something beyond what a bot can provide automatically, the bot quickly escalates the question or topic to a human customer support agent. This relay system ensures that users can still get the assistance they need without unreasonable delay and permits the support team to concentrate their efforts where they are most required.

Overall, Xend AI not only enhances efficiency for the customer support system at Xend Finance but also boosts the general user experience with fast response time and less waiting time. Its capability to filter and handle most inquiries means the human team focuses on less common problems, making support more painless and efficient for all.

The other key tool in use is ticket auto-categorisation. Xend customer support has been improved by adding the automated categorisation of tickets, which has made customer support more organised and quicker. This is particularly helpful for issues that customers report via help tickets. In the past, a customer service agent or engineer had to read each ticket and determine which category the issue related to. These could be things like login issues, wallet errors, or payment issues. This is a time-consuming manual job that slows down customer service.

Today, Xend Finance smart systems with artificial intelligence now do the job. These solutions can automatically triage support tickets by analysing the words used in the message and finding patterns. For instance, it analyses a message having "can't" or "forgot" and categorises it as a ticket related to a login problem. This effectively eliminates the time engineers spend on message triaging. Instead, they can concentrate on fixing problems directly. This system brings many benefits. First, it is a time-saver for the support team. Second, it's a way to make sure the right tickets are assigned to the right engineer responsible for handling these cases. Engineers with particular expertise can now work on tickets that align with their expertise. The result is that customers get help more quickly and more accurately.

And on top of that, the company also has knowledge base recommenders. These are systems that combine a company's database of support articles, manuals and FAQs to suggest the most relevant content when you have a particular problem. When an engineer is talking to a customer, the recommender may suggest an article or a troubleshooting guide based on the context. This is not only time-efficient, but also makes the information shared consistent and accurate. This is also good for new/junior support staff who may not know all the issues, but can still give high-quality support while learning on the job.

All of these AI systems come together to make technical support smoother and efficient. They're not designed to replace the engineers, but to help them. The goal is to alleviate stress, free up engineers to concentrate on work that needs human thinking and creativity, and boost the quality of customer service. However, one of the important questions this research aims to answer is whether these tools truly improve the day-to-day experience of support engineers. For instance, do they feel more productive with these tools? Do the tools make their jobs easier or more complicated? Do they require engineers to learn new technical skills to keep up? These are some of the deeper points that will be explored in the next sections of this study as we look at the real impact of AI-augmented tools or systems on the day-to-day activity of a technical support engineer at Xend Finance.

4 FINDINGS & ANALYSIS

4.1 How AI Is Used In Technical Support At Xend Finance.

This chapter is based on responses from a comprehensive survey that was shared with the staff at Xend Finance. The goal was to see how artificial intelligence tools are being integrated into tech support, what about them is working well and what kind of impact they are having on daily routines. The survey looked at things like how familiar workers are with these tools now, whether they feel their jobs have become more efficient as a result of using them, what the change in job tasks might look like, what new skills will be needed, and how people interact with AI.

What is your current role at Xend Finance?
16 responses

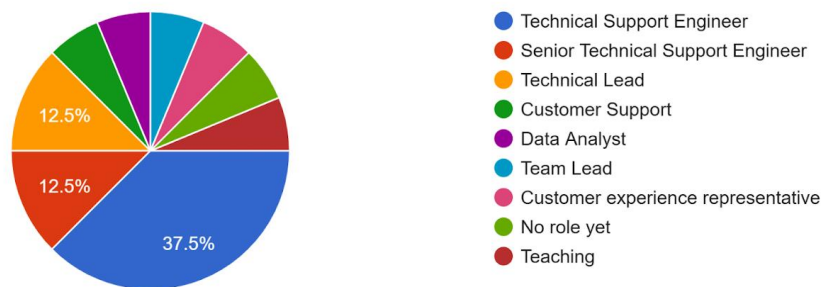


Figure 6: Role of Xend employees who responded to the questionnaire.

17 employees filled out the survey, from different sectors of the company, The responses came from the following categories:

Technical Support Staff (8): This group, which included both engineers and senior engineers, is at the heart of the study since they use AI tools every day while helping customers.

Leads and Managers (4): Team Leads and Technical Leads provided a higher-level view, specifically around training, planning and where AI fits generally in production.

Other Roles (3): A data analyst and a customer experience representative gave feedback on how AI connects with related areas of work, such as data handling and customer service.

Miscellaneous (2): One person said that they are a teacher, another indicated “no role yet.” Responses received from these two were more broad impressions of the company’s use of AI. The findings from the survey show that artificial intelligence is not just a future strategy for Xend Finance but, already an integral part of day-to-day technical support. Employees described how the tools are already being put into action at work, giving an indication for how they’re reshaping tasks and guiding team approaches to customer problems.

One of the most common use cases I've heard is ticket categorisation. Normally, when customer support requests come in, engineers have to read through the details, decide what kind of problem it is, and then assign it to the right category or team. This can be repetitive and time-consuming, especially when there are many cases to process. With the AI system in place, tickets are sorted automatically. This means engineers spend less time organising and more time actually solving problems. Several employees explained that this feature helps them identify urgent cases more quickly and prevents mistakes that may occur when categorisation is done manually.

Another key application is issue prediction and documentation support. Some respondents said the system can recommend what might be causing a problem based on historical data. Alongside that, it often recommends relevant knowledge base articles or troubleshooting guides. This saves support Engineers from having to search through large amounts of information on their own. Rather, they are led straight to potential resolutions, making the support process go much more smoothly.

AI is even being put to use for the automation of repetitive tasks. For instance, it helps to write standard emails and propose steps in troubleshooting or even filling out some of the case details automatically. These small tasks may seem minor on their own, but they accumulate over a busy day at work. Just enough repetition to make manual processing tedious. Automating them frees up valuable time, which engineers can then leverage for concentrating on more complicated and technical problems involving human judgment.

When employees were asked whether AI has improved the way they work, the majority gave very positive responses. Many selected the highest option on the

scale, “strongly agree.” The comments they provided only deepened this conviction. Some said “made my work faster, smart and accurate,” while others explained: “Helps with ticket categorization and suggesting documentation.” A few noted that AI aids them when they are researching cases because it is able to rapidly offer them different potential solutions to try.

The trend is evident from the following chart, where a large majority of the respondents considered AI impactful for their work. This suggests that AI is no longer just an optional add-on but is becoming central to how technical support is carried out within the company.

The introduction of AI-driven tools has improved the way I work.

16 responses

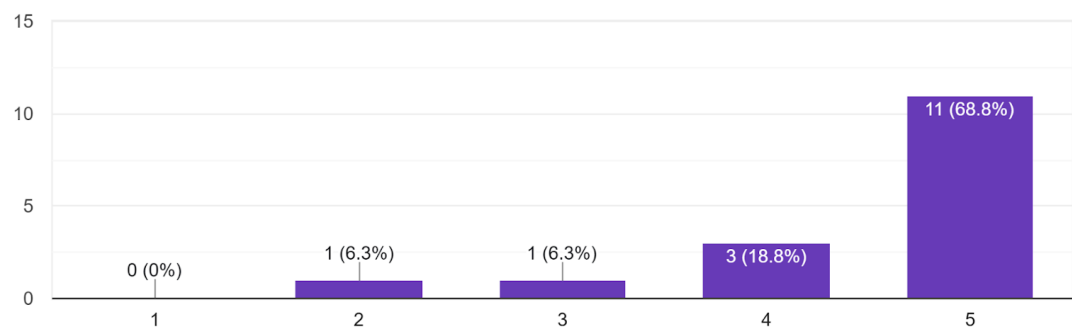


Figure 7: Experience with AI tools in technical support roles at Xend Finance Section

And yet, the responses also indicated that not all workers are reaping the same benefits. While a significant number agreed that AI has enhanced their work, some others were more neutral. That suggests that despite the power of AI tools, employees are still learning to use them or may not be fully comfortable using them yet.

The findings reveal that AI is mainly used for three purposes at Xend Finance: categorising support tickets and predicting issues, linking documentation, and automating repetitive work. Employees generally feel that these tools improve speed, accuracy, and efficiency. However, there is also recognition that the level of adoption and comfort with the tools varies among staff, and more support may be needed to ensure that everyone benefits equally.

4.2 The Impact Of AI Solutions On Technical Engineers

The introduction of AI into technical support at Xend Finance has created clear changes in how engineers approach their work. The survey results reveal both improvements and new expectations that come with the use of these tools.

4.2.1 Productivity

Most respondents agreed that AI tools have increased their productivity. Engineers explained that tasks such as ticket research, case categorisation, and first-level troubleshooting are now completed much faster than before. Instead of manually searching through documentation, AI suggests likely solutions, allowing engineers to spend more time solving the actual problem. For example, one participant shared that “AI makes research easier when solving cases”, while another noted that “it made my work faster, smart, and accurate.” What this demonstrates is that AI is decreasing lag in problem solving and enabling staff to work more efficiently.

The bar chart below from the survey confirms this: the majority of respondents rated 5 (strongly agree) when asked if AI tools help them resolve support cases more efficiently.

I feel that AI-driven tools help me resolve support cases more efficiently.

16 responses

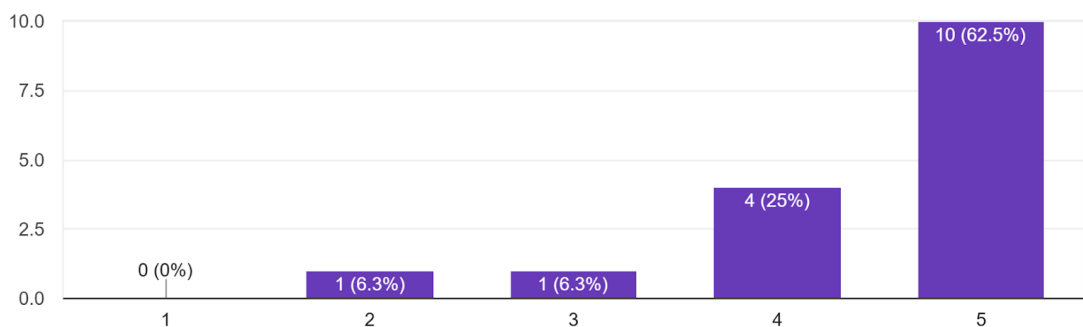


Figure 8: Experience with AI tools in technical support roles at Xend Finance section.

4.2.2 Workload

Another visible effect is the reduction in repetitive workload. Tasks that were previously time consuming, such as email drafting, categorisation, or knowledge base searches, are now partly automated. Engineers reported that this automation frees up time and allows them to focus on more advanced cases that require human judgment and deeper analysis. One respondent mentioned that the introduction of AI has “opened more space to work on complex issues instead of repetitive ones.” This demonstrates how automation is not replacing employees but rather shifting their attention toward higher value tasks.

4.2.3 Job roles

The study also indicates that jobs are slowly evolving due to AI. Technical support is no longer just about answering customer problems. It’s becoming more analytical and technical in nature. The engineers mentioned the need to acquire new skills such as script writing for automation, knowledge of machine learning and using data analytics tools.

I feel that AI requires me to learn new technical or analytical skills.
16 responses

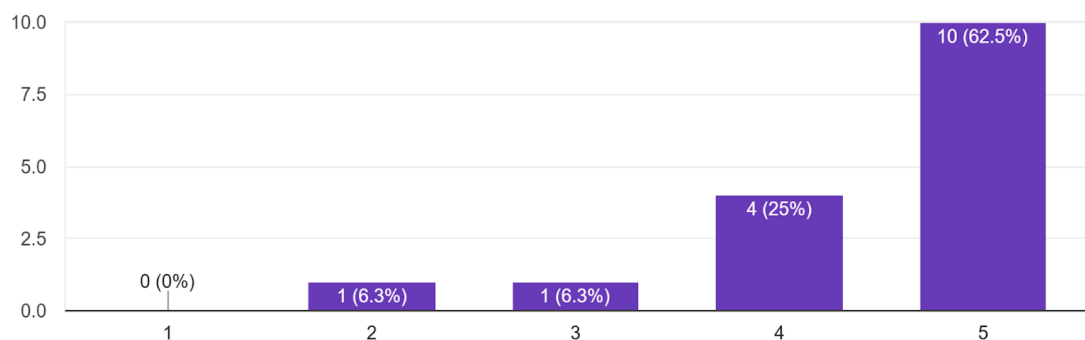


Figure 9: AI impact on roles, skills, and growth section.

Similarly, very few respondents even asked for low code platforms like n8n to be connected into the company Customer Relationship Management (CRM) systems so they could further automate tasks and enhance their use of AI. This demonstrates that AI is changing the expectation of support roles, because they are becoming more integrated with data-driven decision-making and automation.

Since AI tools were introduced, have your job responsibilities changed?

16 responses

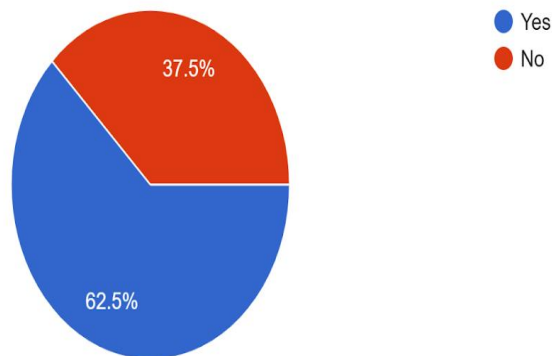


Figure 10: AI impact on roles, skills, and growth section.

4.2.4 Stress levels

When it comes to stress, the findings were mixed. Many engineers said AI reduced stress because it handles repetitive work and provides instant suggestions. This makes their workload lighter and less overwhelming. One participant explained that “AI suggests better troubleshooting ways” which made them feel more supported in their daily tasks. On the other hand, a few respondents admitted that the continuous need to learn new technical skills and adapt to evolving AI tools sometimes adds pressure. Even for those in the industry, the road to mastering AI and knowing how to use it properly can be daunting.

Overall, the evidence shows that AI has brought significant productivity benefits and reduced routine workload. However, it has also reshaped job roles and, for some employees, introduced new sources of stress. This highlights the importance of providing adequate training and support so that productivity gains are not overshadowed by increased pressure on staff.

4.3 From The Employee Perspectives, Do They Find AI Helpful Or Challenging?

AI at Xend Finance is generally seen as helpful rather than challenging. The survey revealed that many staff members view AI as a tool that makes their work smarter, quicker, and more effective.

Several respondents gave specific examples. One person explained that “AI has given me new skills and even contributed to my promotion.” Another noted that “it suggests better troubleshooting steps and makes email writing easier.” These responses show that employees who are comfortable with AI perceive it as an enabler that improves not only their immediate work tasks but also their career prospects.

However, the findings also highlight some challenges. A minority of respondents not familiar with the tools indicate that awareness and exposure to AI is different among employees. Others were worried about the trustworthiness of AI, with some saying that AI could not be trusted blindly. This shows that although AI is beneficial, it needs to be monitored by humans to maintain accuracy and provide relevance. In addition, some participants mentioned the need for more training in order to use the tools effectively.

4.4 Issues Faced In AI Adoption And Change Management

It's clear that there are clear advantages to adopting AI, but the survey also identified a number of blockers pertaining to how seamlessly these tools are incorporated into daily business as usual (BAU). These challenges can be categorised into three segments

4.4.1 Skill Gaps

A lot of the employees acknowledged they have to learn or upgrade their technical skills to keep up with AI adoption. Programming languages, automation tools and machine learning are some of the areas mentioned. Some even mentioned that they needed to improve their basic or overall computer literacy. This indicates

that although AI tools can drive productivity, employees need continued learning in order to effectively utilise them.

4.4.2 Training Needs

In addition to general skills, respondents highlighted the need for structured training and guidance on AI. Several asked for practical, hands on sessions that explain how these AI tools work and what it can realistically achieve. One employee stressed the importance of understanding not only how to use AI but also what its limitations are. Without this kind of support, some staff may feel uncertain or reluctant to trust and adapt these tools in their daily tasks.

4.4.3 Resistance and Implementation Challenges

Some level of resistance was experienced, with some employees stating that AI should not be blindly trusted. Still others had security and compliance concerns, arguing that any tool adopted to support work would have to comply with organisational and industry standards. Some also noted that better compatibility with existing systems like CRMs would make AI more useful. The survey results show that the vast majority of workers acknowledge the importance of upskilling to remain competitive against AI.

This implies Xend Finance should take change management seriously, and invest in training and building trust so that AI adoption is not slowed down by resistance or lack of confidence.

4.5 How Does AI Adoption Affect Job Security And Career Growth?

The last part of the survey explored how employees feel about AI in relation to job security and career development. Interestingly, most participants had a positive outlook. Instead of viewing AI as a threat, they saw it as an opportunity for growth.

Several respondents explained that AI had introduced them to new skills and problem-solving approaches. One person wrote, "I have developed intellectually from the new skills AI brings," while another commented, "Yes, its a lot easier to move to the product development and software development teams now due to the newly acquired technical skills, AI and automation" These responses show

that employees believe AI can help them advance in their careers by making them more skilled and adaptable.

I believe AI tools are opening new opportunities for my career.

16 responses

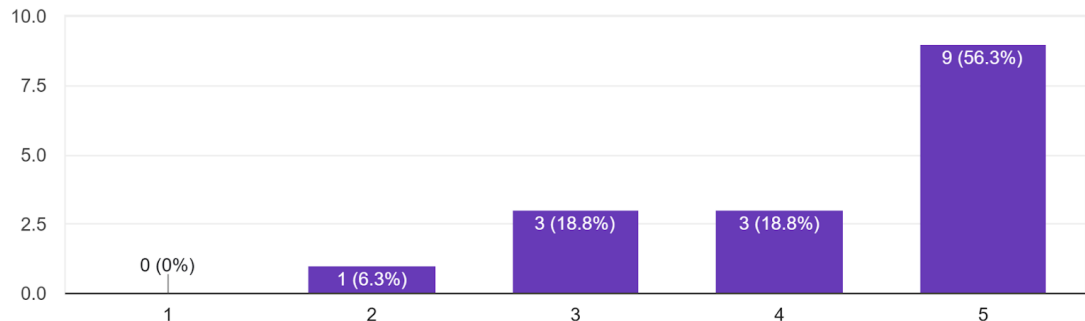


Figure 11: AI impact on roles, skills, and growth section.

The majority of participants strongly agreed that AI tools bring new career prospects. This indicates that, rather than being viewed as a replacement for their function in the organization, employees see AI as a supplementary asset. However, a small number of respondents gave neutral answers, suggesting some uncertainty still remains about long-term effects of AI on job security. In particular, their concern may be that AI's continuing advances might reduce demand for certain routine positions.

In general, though, the results suggest that employees at Xend Finance believe AI adoption supports career development rather than undermines it. To maintain this positive outlook, the organisation should continue to provide training and clear communication about how AI fits into its future plans. This will reassure employees and strengthen their sense of job security.

AI at Xend Finance has been widely adopted and is seen as helpful by most employees. It improves productivity, reduces repetitive workload, and even creates career growth opportunities. However, there are also challenges, including training needs, skill gaps, and occasional resistance. These must be addressed through structured learning and transparent communication to ensure AI continues to be a tool for empowerment rather than a source of anxiety.

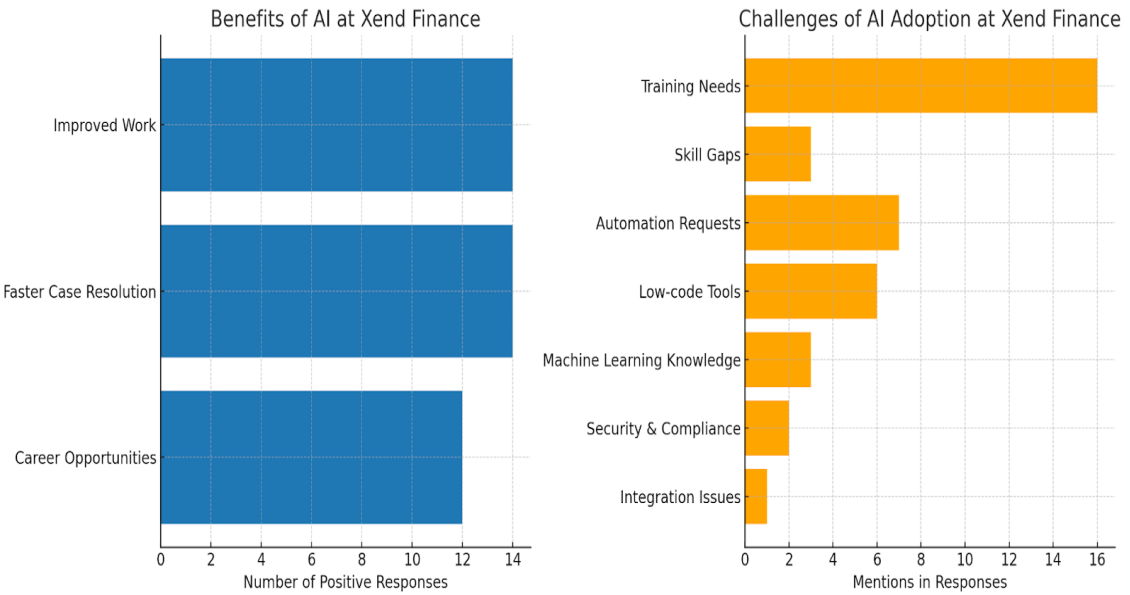


Figure 12: Benefits and challenges of AI adoption at Xend Finance

5 DISCUSSION

5.1 Best Practices For AI Adoption In Tech Support.

The successful implementation of artificial intelligence in technical support is influenced not only by the technology itself but also by how it is introduced, integrated, and managed within an organization. From the experiences at Xend Finance, Numerous practical best practices emerge that can guide other companies seeking to adopt AI tools effectively.

Match AI tools with real operational needs. AI adoption works best when the tools directly address the actual problems that support staff face. If the choices of technology are only made for its novelty or apparent sophistication, then it fails to supply any appreciable benefit. At Xend Finance, employees pointed out that the most practical AI tools were ones that could help in ticket handling, identify where system problems were recurring, and reduce repetitive manual labor. This shows that practical use must receive precedence over technical sophistication. Organizations should first map out where AI has the largest ability to impact on their business and then ensure that the chosen technical aids do satisfy this.

Promote transparency in AI systems, In order for AI to be accepted and trusted by the personnel, transparency is imperative. Employees must understand how the system comes up with its recommendations-which data is used in making them, where its competence ends, as well. Without a clear picture then trust will vanish and adoption slows. At Xend Finance, training courses will not only have taught staff how they should operate the AI tools but also showed them how to interpret outputs and judge their dependability. This involved explaining the underlying logic and pointing out potential error scenarios.

Also, best practices will involve Introducing AI gradually, in stages. The gradual adoption is more effective than implementing AI all at once. At Xend Finance, the company began by integrating AI into specific, manageable tasks, such as ticket triage and preliminary system log analysis, before expanding its scope. This step-by-step approach lets engineers change gear at their own pace, and management can catch problems or clogs early. A rolling schedule in this way will also reduce resistance, with the consequence that employees are more inclined to

pick up techniques incrementally and thus can be persuaded to learn almost anything.

Establish continuous feedback loops to boost AI adoption. This should be treated as an ongoing process rather than a one time event. Employees must be able to provide feedback on the performance of AI tools, including whether they help, create new challenges, or require improvement. At Xend Finance, engineers were encouraged to report system errors, suggest enhancements, and share lessons learned from daily interactions with AI. This feedback not only improves tool effectiveness but also fosters a sense of ownership among employees, making them active participants in the transformation rather than passive users.

Provide comprehensive training and support. Beyond simple instructions, AI adoption is most successful when staff receive detailed, practical training that covers both operational use and problem-solving with AI. Xend Finance Engineers emphasized hands on workshops, scenario based exercises, and real case simulations. These sessions will enable employees to experience AI in contexts similar to their daily work, strengthening both competence and confidence. Ongoing support, refresher courses, and accessible resources further help staff stay up to date as tools evolve.

Fostering a collaborative culture around AI, a successful AI adoption requires a supportive cultural environment as well as preparing technically. Employees need to be able to question AI outputs, suggest improvements and work with both peers and management toward maximizing workflow efficiency. At Xend Finance, team discussions and peer sharing sessions will enable the staff to exchange thoughts, gain something from other people's mistakes and carry out strategies for how AI would be integrated into operations. In such an environment of mutual learning withholding judgments about errors, ideas progress smoothly without resistance.

Furthermore, Align AI adoption with broader organizational strategy. AI tools should not be implemented in isolation, They must be connected to the company's larger strategic objectives, such as improving customer satisfaction, streamlining operations, or enhancing competitive advantage. At Xend Finance,

AI served as part of the general quest for enhanced operating efficiency combined with excellent customer support. Aligning AI efforts with business objectives guarantees that the technology delivers measurable results in conjunction with other management tools rather than becoming simply isolated experiments.

Adopting AI in technical support is more than just implementing new software. It involves careful planning, employee engagement, incremental deployment, transparent communication, and ongoing support. Organizations that combine these practices with cultural readiness and strategic alignment, as demonstrated by Xend Finance, are more likely to achieve successful and sustainable AI integration that enhances both employee performance and overall business outcomes.

5.2 How To Balance AI and Human Expertise.

One of the key lessons that emerged from the Xend Finance study is that artificial intelligence should not be treated as a substitute for human knowledge, but rather as a supportive partner. Striking the right balance between people and technology is what determines whether AI adoption improves results or ends up creating confusion and frustration. Artificial intelligence has many advantages in performing quickly, recognizing patterns, and handling routine tasks.

For example, it can scan the logs of systems in seconds, find recurring customer issues, and even provide answers derived from previous cases for such helpdesk questions. This kind of automation relieve technical team members to concentrate on higher priority tasks and save time. Yet there are also disadvantages in employing AI. Sometimes they give incomplete answers, misinterpret things that only a person can catch through experience. That is why the human element remains so important.

At Xend Finance's several engineers of the support department reported that AI had indeed eased the burden from the daily handling of tickets. Instead of spending hours filtering out repetitive cases, they were now able to give more time to complex or non-standard problems. But their role still did not disappear, it turned towards reviewing, interpreting and if necessary correcting AI system suggestions.

In this way, the best results came when mechanised efficiency combined with human insight and judgement was used. To achieve a happy medium between human input and AI, Organisations must build a culture in which engineers feel confident to question AI generated results by challenging or double checking this recommendation should they seem unreliable. If engineers rely heavily on AI, errors may slip through unnoticed, yet the lack of trust in AI can leave potential untapped. To avoid extreme mistakes, management should remind employees that AI is designed to assist them, not to decide their final choices.

Another important perspective for achieving equilibrium is having an understanding of the differing roles each can play. AI excels at such tasks as maintaining course, speedily processing information and intuiting information from data, yet it lacks the ability to be creative, empathetic or context-sensitive. People, though, can deal with the unexpected, grasp a customer's emotions and think in strategic terms, areas in which machines simply cannot compete. This insight into their complementary strengths means a proper assignment of AI and human expertise to tasks as called for.

Communication with customers also demonstrates the importance of balance. While AI chatbots or automated systems may provide quick responses to routine questions, many customers still prefer the security and warmth to a skilled human ear when their problem is both serious and sensitive. This approach will help with serving two purposes better. Both efficiency and customer satisfaction will be improved. Balance is not something that will last, It requires continuous reassessment and nurturing as tools evolve, and employees gain experience using them. Ongoing training sessions, clear role definitions, and feedback collection can help maintain this delicate equilibrium.

Companies that succeed are the ones that consistently reexamine how work is divided between man and machine and adjust the balance of roles accordingly. In other words, the most effective approach is to see AI as an assistant that makes work easier, not as a replacement for human expertise. At Xend Finance, this perspective will allow engineers to make better use of their time, focus on complex thinking, and still maintain ownership of the decision-making process.

The lesson is clear for when people and technology work together in a complementary way, the overall quality of technical support improves significantly.

5.3 Ways To Improve AI-Related Training Programs.

In AI, One of the most important revelations of the Xend Finance survey is that training is critical to how well employees make use of new technology. While a growing number of staff were able to operate AI tools provided at worksites, most acknowledged that they could not fully understand how these systems reached their conclusions or what their weak points might be. This gap underscores what is required, not just basic demonstrations in training programmes, but also to create self-confidence, sense of inquiry and long-term handling capacity.

A good starting point is to cover the fundamentals. Employees should know what the tool does, how it processes data, and the types of mistakes that can occur. By explaining the logic behind the system, training reduces blind trust and prevents workers from assuming that the machine is always right. When people are aware of possible errors, they can approach outputs with a healthy level of caution. This idea reflects what we discussed earlier that organizations must prepare their workforce to use new digital technologies effectively, not just introduce the tools (Hess et al., 2016).

Another important improvement is training according to role. Different teams use AI in different ways, so one single, general session often leaves out large missing pieces. For example, technical staff might need detailed study on how to detect errors and shape work flows with AI help while managers on the other hand will be more concerned with keeping the system operating well, tracking numbers, and resourcing decisions. Everyone gets tailored training specific to their job responsibilities. This view is consistent with (Bharadwaj et al. 2013), who argue that successful digital transformation requires aligning people, processes, and technology according to the specific needs of different groups.

Practical experience is equally important. Employees learn best when they can apply knowledge to real life scenarios. Workshops, live demonstrations, and case-based exercises allow staff to practice using AI in situations that closely

resemble their day-to-day tasks. At Xend Finance, sessions that allowed employees to work with actual ticket data, classify issues, and test automated solutions were reported to be more helpful than lectures or manuals. This hands on style of training builds familiarity and reduces resistance. Westerman, Bonnet, and McAfee (2014) describe this as combining digital capability with leadership capability, workers must be given not only the tools but also the confidence to use them productively.

Training should be an ongoing process once people have started to learn well. Technologies do not stand still, and the learning around them must keep up with their changes. Refresher courses from time-to-time, short updates occasionally, and opportunities provided by peer learning help employees keep pace with developments.

Some Xend Finance employees mentioned they would gain more from brief, ongoing sessions where colleagues could share tips or demonstrate new functions they had found.

This type of peer learning is less formal but often more effective. In addition, training programs have to encompass the ability to think critically about what employees are being taught. Not only should they learn how to pull the levers and turn the knobs, but also what their machine spits out and escalate cases when results look unreliable, and report errors back to system administrators. Training that combines technical know-how with critical thinking prepares employees to act responsibly rather than blindly following the machine.

Also, to improve training is to provide employees with resources they can return to after the initial sessions. With such support, employees can continue learning and dig deeper into key points. Step-by-step guides, online tutorials, and internal knowledge bases to ensure the learning goes on even while employees are at work. When necessary, employees can search for the instructions they need in real time.

Finally, leadership involvement is vital to the success of training. For example, if managers encourage attendance, give some time for learning and display an interest in employees' feedback, training turns into something rather more like an investment than a chore. Employees at Xend Finance thought that leadership

involvement would make the practice more credible to them, since top level commitment is evident in every aspect of operations throughout an organization.

Training Employees on AI needs to be workable, tailored and continuous. It should not be about the 'how' but also the 'why', and employees need plenty of opportunities to practice in natural settings. Nor must the lesson stop once learned, training must be reinforced over time. By regarding training as a continuous partnership between employee, trainer and leadership, companies like Xend Finance can make sure that staffs continue to feel confident, able and ready for digital trends as they come along.

5.4 Challenges in AI-human collaboration

While the integration of AI tools in technical support is very advantageous, the Xend Finance case study also indicates that cooperation between people and computers is no bed of roses. In most cases these difficulties are not directly related to technology itself but rather to how people communicate with it. Understanding these issues is key to ensuring that the adoption of AI really does turn out as intended, raising productivity and not creating new frustrations.

One major problem is trust and reliability. What many colleagues notice about artificial intelligence is that although it often provides useful suggestions, it does not always give the correct or complete answer, and a small mistake in technical support may confuse an engineer, or it may make a customer upset as well, sometimes both. When this happens, staff often hesitate to rely upon the system completely. It can be a frustrating experience for recent joiners who lack confidence in their own judgment to balance the trust we place in AI recommendations against our human intuition. As (Hess et al. 2016) point out, digital transformation succeeds only when employees feel supported in adapting to new tools rather than pressured to accept them blindly.

A second challenge is shifting job roles and responsibilities. At Xend Finance, support engineers observed that AI automated many repetitive tasks, such as sorting tickets and scanning system logs. While this freed up time, it also changed what was expected of them. Their role increasingly involves interpreting AI out-

puts, checking for errors, and handling more complex issues that machines cannot solve. This shift can be empowering but also unsettling, as some employees feel unprepared for higher-level analytical tasks. As (Bharadwaj et al. 2013) mentioned that such changes require organizations to align people, processes, and technology in ways that support continuous learning and adaptation.

Another problem is overdependence. When staff rely too heavily on AI tools, they can easily become accustomed to having others solve their problems for them. For example, one respondent from Xend Finance mentioned that newer team members often defaulted to AI recommendations without taking the time to verify them. This leads not only spreading mistakes but also weakens the engineer's own diagnostic abilities over time. Guarding against this pitfall needs a culture that promotes inquiry, independence of thought and a readiness to overrule machine recommendations if necessary.

Communication gaps can also cause problems. AI systems typically present outputs in technical language or provide simple suggestions, but these may not be understood by staff in all cases. If explanations are unclear, employees may misunderstand what the system is recommending or misapply its results. This highlights the importance of designing AI tools with transparency and user-friendliness in mind. As (Westerman, Bonnet & McAfee 2014) suggested that, technology must be paired with strong leadership and clear communication to translate digital potential into business value.

A third hurdle comes with training and trust in that training. The Xend Finance employees who used the tools knew that they worked but were not always entirely clear on how. Such lack of knowledge makes it more difficult to recognize when a system is wrong, leaving itself open to two kinds of error, either completely blind faith or overtly doubting everything. Training programs which concentrate not only on technical competencies but also critical evaluation, as outlined previously, are necessary to overcome this barrier.

Emotional and cultural factors also have an impact. For some employees, AI may represent a threat to their professional identity, leading them to fear that increased automation will make their expertise less valuable. Others feel overwhelmed by the pace of technological change. These sentiments are potential sources of resistance even when they cannot point to any objective harm caused

by using modern digital tools. Therefore, not just the technical aspect but also the human must be considered in AI adoption efforts. Employees need to be made aware that despite all these innovations, their expertise remains central to the support process.

Overall, the challenges of AI and human collaboration are multi layered, ranging from technical reliability and training needs to cultural acceptance and role adjustment. The Xend Finance experience demonstrates that while AI can streamline workflows, it cannot replace the judgment, creativity, and contextual understanding of human engineers. So to make collaboration work, organizations must actively manage trust, encourage continuous learning, and support employees through the cultural shift that digital transformation demands.

5.5 Conclusion / Recommendations

Based on the findings from Xend Finance and insights from digital transformation Theory, several recommendations can guide how organizations can better adopt AI in technical support. These recommendations address not only the technical aspects of adoption but also the human, cultural, and organizational sides.

Strengthen training and continuous learning, the survey results made it clear that while employees could use the tools, many lacked confidence in understanding how they worked. To address this, companies should provide training that is role specific, practical, and ongoing. Engineers need technical depth, while managers need strategic insights. Learning should not be a one-time event but part of a continuous process. Refresher courses, peer-to-peer learning, and simulation-based exercises will ensure employees remain capable as tools evolve.

Position AI as a partner, not a replacement. Xend Finance employees described AI as making their work lighter, but also having the effect of transferring responsibilities towards more complicated tasks. To make the transition smoother, organizations need to position AI as an assistant that complements human expertise. Leadership can reinforce this by encouraging staff to challenge AI recommendations, ensuring that human judgment remains central. As (Westerman, Bonnet, & McAfee 2014) argued that, digital masters succeed by combining technology with strong leadership, not by substituting one for the other.

Build trust through transparency and reliability. A recurring concern from the survey was uncertainty about how reliable the AI tools were. Organizations should make AI systems more transparent by showing how recommendations are generated and by clarifying possible limitations. Trust can also be strengthened by creating feedback loops where employees can report errors or inconsistencies, which are then used to improve the tools. This turns AI adoption into a collaborative process rather than a top-down imposition.

Support employees through role transitions. Once AI starts taking on routinized responsibilities, employees have to move into roles focusing more on analysis, decision making skills and creativity. This change may be empowering. Yet it is also likely to leave many people apprehensive and without guidance. Managers need to both communicate clearly just what exactly new expectations are in this environment, help staff acquire different competence areas along with old ones and recognize that staffs bring the human touch and add value to machine efficiency. This also reflects the message of (Hess et al. 2016), that digital transformation requires strong organizational support for employees.

Foster a culture of questioning and innovation. AI adoption is most likely to succeed when staff are brave enough to question its results and daring in proposing amendments. Organizations should nurture an atmosphere where employees will share their experiences, point out weak points in the tools and even suggest new applications.

Align AI adoption with broader business strategy. Finally, In the end, organizations need to avoid seeing AI initiatives as mere isolated projects but make sure they are linked with long term objectives of efficiency, customer satisfaction and innovation. In fact, handling tickets and workflows will be improved with AI, but senior management should also link these achievements to broader aims like improved customer experience or raising the standard for rival products. When AI adoption is tied to strategy, the impact sticks. It should not be a breakthrough today only to be forgotten tomorrow. Instead, technology attains lasting effects on the organization. Also, The successful use of AI in technical support is not just a question of new software. It means training, changing corporate culture, building trust and getting strategies right.

5.6 Summary

Whether or not AI can be applied within technical support is both a great opportunity and huge challenge at the same time. In the Xend Finance story, we can see that AI tools not only make efficiency and save work on routine tasks too quick and accurate. They also change the role of technical support engineers in a way that calls for new types of skills from them. Those working with such novel ways need greater flexibility than in former periods. It is clear that AI is not a replacement for human expertise. Instead, its greatest value comes when it is combined with human judgment, creativity, and contextual understanding.

Engineers at Xend Finance reported that while AI could quickly analyze logs and suggest likely solutions, their role remained essential in verifying results, solving complex cases, and ensuring that customers received accurate and empathetic support. This reflects the essence of Digital Transformation Theory. (Westerman, Bonnet, & McAfee, 2014), which emphasizes that organizations must combine technological and leadership capabilities to truly capture digital opportunities.

At the same time, the challenges identified in this study such as lack of trust, shifting job expectations, and insufficient training highlight that digital transformation is as much a human process as a technical one. Dynamic Capabilities Theory (Teece, Pisano, & Shuen, 1997) helps explain why organizations must continually adapt by reallocating resources, retraining employees, and fostering new ways of working in order to sustain competitive advantage in a changing environment.

The key lesson is that AI adoption in technical support must be approached holistically. It is not enough to introduce a new tool, employees need to be supported, roles must be redefined, and a culture of questioning must be encouraged. Leadership plays a decisive role in ensuring that employees see AI as a partner rather than a threat, and that they are equipped to work confidently alongside these systems. Ultimately, the Xend Finance experience shows that AI can reduce mechanical tasks, free up time for more meaningful work, and enhance the overall customer experience. However, its success depends on how well organizations balance human expertise with technological capabilities. If this balance is achieved, AI will not only transform technical support but also strengthen the organization's long-term ability to innovate and compete in the digital age.

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APPENDICES

Appendix 1. The Questionnaire

Section 1: Background Information

What is your current role at Xend Finance? *

- ☐ Technical Support Engineer
- ☐ Senior Technical Support Engineer
- ☐ Technical Lead
- ☐ Other: _____

How long have you worked at Xend Finance? *

- ☐ Less than 1 year
- ☐ 1–2 years
- ☐ 3–4 years
- ☐ Over 4 years

How familiar are you with the AI-driven tools used at Xend Finance (e.g Xend AI chatbot, auto-categorisation, issue prediction)? *

- ☐ Not familiar
- ☐ Somewhat familiar

- ☐ Senior Technical Support Engineer
- ☐ Technical Lead
- ☐ Other: _____

How long have you worked at Xend Finance? *

- ☐ Less than 1 year
- ☐ 1–2 years
- ☐ 3–4 years
- ☐ Over 4 years

How familiar are you with the AI-driven tools used at Xend Finance (e.g Xend AI chatbot, auto-categorisation, issue prediction)? *

- ☐ Not familiar
- ☐ Somewhat familiar
- ☐ Familiar
- ☐ Very familiar

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Experience with AI Tools in Technical Support Roles at Xend Finance

Kindly answer the questions in these sections using the score ratings between 1-to-5 with 1 meaning that you have no idea of the subject asked and 5 meaning that you have full knowledge of the subject.

The introduction of AI-driven tools has improved the way I work. *

	1	2	3	4	5	
Strongly disagree	☐	☐	☐	☐	☐	Strongly agree

If you agree or disagree give at least one example (or instant) *

Your answer _____

I rely on AI-driven tools like 'auto-categorization and issue prediction' in my daily tasks. *

	1	2	3	4	5	
Strongly disagree	☐	☐	☐	☐	☐	Strongly agree

I feel that AI-driven tools help me resolve support cases more efficiently. *

	1	2	3	4	5	
Strongly disagree	☐	☐	☐	☐	☐	Strongly agree

Can you share situation(s) where this was true or not true? *

Your answer _____

I understand the purpose and goals of using AI-driven tools at Xend Finance. *

	1	2	3	4	5	
Strongly disagree	☐	☐	☐	☐	☐	Strongly agree

Do you participate or involved in introduction of new AI tools. *

Example: involved in the research and evaluation of tool options down to testing the integration of such tools

	1	2	3	4	5	
Strongly disagree	☐	☐	☐	☐	☐	Strongly agree

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Benefits and Challenges of AI tools in Technical Support roles

Please select more than one option

What are the biggest advantages you have experienced from using AI tools in your support work? *

- ☐ Faster ticket handling
- ☐ Better prioritization of issues
- ☐ Access to helpful recommendations
- ☐ Reduced manual workload
- ☐ Improved customer satisfaction
- ☐ Other: _____

What challenges or difficulties have you faced when using AI tools? *

- ☐ Tools make wrong suggestions
- ☐ Lack of training or documentation
- ☐ Difficulty in trusting automated decisions
- ☐ Too many system notifications or suggestions
- ☐ Other: _____

Do you feel that AI has made your job easier or more difficult? *

Your answer _____

AI tools have affected the way I approach problem-solving. *

	1	2	3	4	5	
Strongly disagree	☐	☐	☐	☐	☐	Strongly agree

Can you describe how? *

Your answer _____

I believe AI tools are opening new opportunities for my career. *

	1	2	3	4	5	
Strongly disagree	☐	☐	☐	☐	☐	Strongly agree

Please explain *

Your answer _____

Suggestions and Final Thoughts

In your opinion, what can be done to improve the use of AI tools in Technical Support at Xend Finance? *

Your answer

How do you think humans can work best together with AI tools in day-to-day tasks? *

Your answer

What support or training would help you feel more confident when working with AI tools? *

Your answer

Any final thoughts, feedback, or suggestions regarding the use of AI in your work? *

Your answer